

4DIP: A Self-Stabilizing Framework for Predicting Physical Systems

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https://github.com/ecrurefuse/curry_4DIP

Abstract

This paper introduces the 4DIP framework (4-Dimensional Iterative Prediction), a general-purpose simulation method designed to evolve physical systems while naturally removing singularities and capping divergent infinities. 4DIP integrates two independent convergence rules—one for energy and one for time—that allow the system to automatically slow down in sensitive regions and speed up in stable ones. The result is a numerical method that does not rely on externally imposed tolerances, empirical error bounds, or artificial regularization.

Rather than simulate systems for fixed durations or fixed step counts, 4DIP uses symbolic damping based on the instantaneous change in energy, allowing the system to stop when physical quantities stabilize to high precision (e.g., change in energy $< 10^{-14}$). This self-regulation enables both predictive modeling and singularity removal without violating underlying physical laws.

The framework is validated on a series of vector, scalar, and tensor-based physical equations including ordinary and partial differential equations. In all cases, 4DIP successfully converges without requiring any tuning, using only symbolic physics, dynamic damping, and adaptive time logic. Additional applications, including chaotic systems and relativistic orbital dynamics, are reserved for follow-up publications.

1 Introduction

In modern physics, some of the most important equations break down exactly where we need them most: at small scales, high densities, or near field singularities. Standard tools such as Runge–Kutta solvers, Euler methods, or even lattice-based techniques offer powerful ways to evolve physical systems—but they do not fundamentally address the mathematical instabilities caused by infinities or the absence of built-in physical convergence criteria.

Singularities appear in both classical and quantum systems. For example, the electric field from a point charge diverges as $r \rightarrow 0$, and curvature becomes infinite near black holes at sub-Planck scales. These breakdowns are often treated by regularization, cutoff parameters, or statistical averaging—techniques which may work but are mathematically artificial and physically opaque.

4DIP was developed to solve this problem at the source. Rather than impose fixed tolerances or pre-defined simulation lengths, 4DIP evolves a system until it naturally stabilizes in energy and time. The framework consists of a simple iterative rule that combines symbolic damping with adaptive time step scaling, enabling systems to stop themselves as soon as they reach a physically meaningful state.

Most importantly, 4DIP is constructed entirely from first principles. It does not require empirical adjustment, does not rely on numerical smoothing, and does not assume convergence. Instead, it ensures that systems reach equilibrium—or remain bounded—based on the very quantities they’re simulating.

This paper focuses on presenting the updated 4DIP framework, refining the original method into a form that can be tested on real hardware, validated against reference physics, and used to simulate a wide range of scalar, vector, and tensor-based systems. Although chaotic systems and orbital dynamics are left for subsequent publications, the results in this work show that 4DIP is already a viable alternative to conventional numerical solvers, especially for high-precision physical applications.

2 Methodology

Predictive Function

The core of the method begins with a predictive function:

$$F_n = G_n + \frac{dG}{dt} \cdot \Delta t_n$$

where F_n is the forward-predicted state, G_n is the current system state, and Δt_n is the timestep at iteration n .

State Update (with and without damping)

Without damping:

$$G_{n+1} = G_n + f(G_n, t_n) \cdot \Delta t_n$$

With symbolic damping:

$$G_{n+1} = G_n + (f(G_n, t_n) - \eta_n G_n) \cdot \Delta t_n$$

Symbolic Damping (Energy Control)

$$\eta_n = \max\left(0, \frac{dE/dt}{E + \delta}\right)$$

where δ is a small constant to avoid division by zero. This damping slows the system as its energy stabilizes.

Time Control (Fourier-Scaled Logic)

$$\Delta t_{n+1} = \Delta t_n \cdot \exp(R_n - 0.5)$$

$$R_n = 0.5 + 0.5 \cdot \exp\left(-\frac{\|F_n - G_n\|}{\|G_0\|} \cdot \cos\left(2\pi \cdot \frac{T_{\text{final}} - t_n}{T_{\text{final}}}\right)\right)$$

Termination Criteria

- Time-based: $t_n \geq T_{\text{final}}$
- Energy-based: $|\Delta E_n| = |E_n - E_{n-1}| < \epsilon$

Precision Tiers

Tier	Threshold ϵ	Application
1	10^{-6}	Engineering, mechanical models
2	10^{-8}	Sensors, electromagnetic equations
3	10^{-10}	Orbital mechanics, astronomy
4	10^{-12}	Nanoscale and wave systems
5	10^{-14}	Quantum convergence (used here)
6	10^{-16}	Symbolic/special-purpose (float128)

Table 1: Precision convergence tiers in 4DIP

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